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Industrial Attachment Report

Major Group: Electrical Energy and Power System (EEPS)

Industrial Attachment Report: Ashuganj Power Station Company Limited (APSCL)

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1. Abstract

Ashuganj Power Station Company Limited (APSCL) stands as a cornerstone of Bangladesh's energy infrastructure and a successful model of the government's Power Sector Development and Reform Program. Established as a public limited company following its transition from the Bangladesh Power Development Board (BPDB) in 2003, APSCL has evolved into a key driver of the national grid. With an installed capacity of 1,647 MW, the company leverages its strategic location near the Titas Gas Field and the Meghna River to convert indigenous natural gas into bulk electricity, fueling the country's industrial and domestic growth.

Beyond power generation, APSCL serves a dual role in regional development and environmental resource management. The water utilized for plant cooling and steam generation is discharged into irrigation channels during the dry season, supporting approximately **40,000 acres of farmland** in Brahmanbaria and surrounding districts. This integration of industrial output and agricultural support highlights APSCL's multifaceted contribution to the national economy.

Looking toward a sustainable future, APSCL is aggressively modernizing its fleet with energy-efficient, gas-based technologies while diversifying its energy mix. The company is currently expanding its footprint into **renewable energy and imported LNG-based projects**, notably through land acquisition in Patuakhali and solar initiatives in Narshingdi. By balancing technological innovation with social responsibility, APSCL remains a vital institutional asset in Bangladesh's journey toward energy security and long-term economic prosperity.

2. Introduction

Ashuganj 420MW Gas Power Plant (EAST) is a Combine Cycle Power Plant (CCPP). Since the plant itself requires about 5% of its generated power to operate, the installed capacity and the present capacity of the power plant is 400MW and 393MW respectively. [PGCB NOTE](#)

Ashuganj 450MW CCPP (North) has installed capacity of 360MW and present capacity is 353.35MW. This plant is connected to 400kV towards Bhulta through Gas Insulated Substation (GIS) and 230kV bus through two 325MVA transformers. [MARCH 2025 BD power grid](#)

The configuration of this GSUT 20 BAT 10 suggests the Bus Auxiliary Transformer Rated at 10MVA while generating 20kV.

Ashuganj 450MW (South) has the same installed capacity as North project. This plant can generate 318.81MW if necessary. This plant generator produces about 22kV and the GSUT supplies power to the 230kV bus. The compressor of GT was under maintenance during industrial attachment visit.

Ashuganj CCPP 225MW plant has two different generators for the GT and the ST. The multi shaft or, split shaft design ensures. One generator has installed capacity of 142MW while the other one is 75MW. The plant can generate up-to 221MW. Both of the generators are connected to 132kV bus through GSUT. The GSUT is rated at 11kV/132kV.

3. Departments

3.1. Monitoring

To observe, to check the conditions of the received values from transmitters are need to be read. The operator then decides what should be done. In control center, the position of each valve, pneumatic pressure valve, oil pressure valve position, the temperature of key locations, are generally monitored all day long. No abrupt changes are allowed. The operators try their best to run the plant under optimal condition. The operator knows how much importance it carries if the temperature of steam is less than 85°C after low pressure steam turbine.

The operator monitors the pressure and the temperature of High-Pressure boiler, CRH boiler and bypass valve all day long to run High Pressure steam turbine (HP ST). The operator must take into measure the temperature and pressure of low pressure (LP) boiler and also for intermediate pressure (IP) boiler. Thus, the operator decides whether the steam turbine is running efficiently or not. Otherwise, the most important part of CCPP, heat recovery steam generator (HRSG) won't be that much of effective device to meet certain demands. Same goes for air compressor. The pressure ratio of 1:18 won't be maintained and the

gas turbine (GT) won't get to the state of generating required amount of power. The operator also monitors close cycle water (CCW) temperature and open cycle water (OCW) temperature to get an idea of system transparency. The operator must take into account how much load is to be connected to the system and how much real power to be generated along with how much reactive power is to be injected to the system to deliver power but considering high efficiency. The monitoring system has some special parameters known as fail proof parameters that are set by the operator once and being monitored automatically to ensure the process continuity. The control system then energizes the actuator to meet the demand.

3.2. Controlling

Different types of sensors are placed at different positions of the devices in a power plant. The sensor can immediately sense the difference in condition and the signal needs to be sent to the operator. So, a transmitter device is set after sensor. This transmitter device structurally consists of a sensing probe that is being used to take sample, a cable duct to send the signal back to the operator. The operator then decides whether to react or not to the received data. Some special parameters are also present there which need to be ensured at recommended state. To ensure such condition fail proof system is integrated. This data is to be achieved by some supervised condition to meet optimal running condition. This part of controlling is always automated but the operator might set those parameters depending on the condition of power plant. The implementation of control system is done by actuator devices. Actuators are the devices that receives a signal from system and acts upon it by setting a new position of valves. Power plant has different types of valves. Pneumatic valves, the oil pressure or gas pressure, steam pressure valves and so on are present in a power plant. Some devices actuate upon spinning of motors that actually sets the position of valve. These sensitive devices act upon slight changes in values and to ensure their smooth and simultaneous working, the actual high pressure or high-power valves are set to act upon second stage of decisions known as cascaded valve operation.

For example, a type of actuator acts when the diaphragm is pressed but the isolated part senses the small change of condition and then it energizes the actuator to implement the operation.

The electrical part designed some local control box so that the operator can control any device locally if necessary while the system is fully functional and automated. That's why, the junction box and local control box are implemented.

3.3. Maintenance

Maintenance department ensures the machinery, equipment and system operate efficiently by performing preventive maintenance, diagnosing and repairing issues, and implementing improvements to minimize downtime. They are responsible for tasks like scheduling repairs, managing spare parts, overseeing maintenance budgets, and sometimes managing a team of technicians.

Transformer Maintenance Overview

Maintenance of power transformers is a systematic process designed to monitor the health of the unit and prevent catastrophic failures. It is generally divided into Routine (Scheduled) maintenance and Condition-Based monitoring.

1. Routine Preventive Maintenance

These tasks are performed at fixed intervals (Daily, Monthly, or Yearly) to ensure all auxiliary components are functioning.

- **Daily Inspection:** Checking oil levels in the conservator, monitoring load current and voltage, and verifying operating temperatures (OTI/WTI).
- **Silica Gel Breather:** Inspecting the color of the silica gel. If it turns from blue to pink, it indicates moisture saturation and must be replaced or regenerated.
- **Cooling System Check:** Verifying that cooling fans and oil pumps are operational and free of unusual vibrations.
- **Cleaning:** Removing dust and contaminants from bushings to prevent tracking and flashovers.

2. Oil Maintenance & Management

The insulating oil is the "lifeblood" of the transformer. Its maintenance is critical for cooling and insulation.

- **Dielectric Strength Test (BDV):** Testing the oil's breakdown voltage to ensure it can withstand electrical stress.
- **Filtration/Dehydration:** If moisture or sludge is detected, the oil is passed through a filtration plant to remove impurities and dissolved gases.
- **Gasket Inspection:** Checking for oil leaks at valves, bushings, and tank joints to prevent environmental hazards and moisture ingress.

3. Periodic Testing (Shutdown Maintenance)

Performed during scheduled outages (Annual or Bi-Annual) to assess the internal integrity of the unit.

- **Tightening Connections:** Ensuring all electrical terminals (HV/LV) are tight to prevent localized overheating.
- **Relay Testing:** Manually triggering the Buchholz relay and Pressure Relief Device (PRD) to ensure the protection circuit correctly trips the circuit breaker.

Insulation Resistance (Megger): Measuring the resistance between windings and the ground to detect insulation aging.

3.4. Instrumentation and Control (I&C)

Instrumentation and control department designs, develops, and processes variables like pressure, temperature, humidity, flow, pH, speed. They combine knowledge of electrical and electronics. The automated system consists of PLC cards, relays, contactor, VSD.

The engineers are supposed to be expected to be at field because they had so many to ask. To summarize the situation, in times of necessity, the engineers are expected to be present until the problem is solved. The NORTH CCPP electrical design shows some protection panel acting on equipment safety protocol and some protection panel working on 400kV line protection. The automated control system shows how modern power plant facilities ensure great safety to human kind. To check the pressure of condenser, a diaphragm mechanism is implemented and the other partition of diaphragm senses the changes of pressure and thus the transmitter works in real time. In some cases, majority rule is applied to ensure data accuracy and so 3 transmitters are placed to sense the data. Majority rule ensures whether the data is correct or not.

3.5. Simulation Model

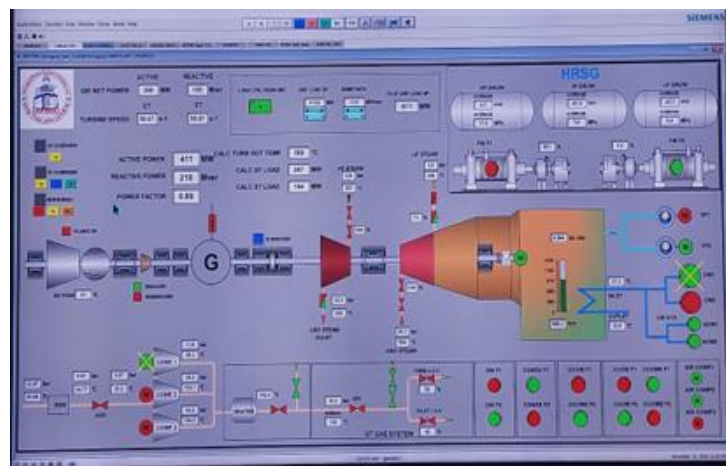


Fig. 1: DCS System Overview

4. Substation Grid

In modern power systems, the choice between AIS (Air Insulated Substation) and GIS (Gas Insulated Substation) is one of the most critical design decisions. While both perform the same fundamental task—switching and protecting the grid—their physical construction and insulation methods are vastly different.

4.1. Substation Types

4.1.1. Air Insulated Substation (AIS)

In an AIS, the primary insulation between the high-voltage conductors and the ground (as well as between phases) is ambient air.

Design: Equipment like circuit breakers, isolators, and busbars are installed in the open air. Because air has relatively low dielectric strength, large clearances (distances) are required to prevent electrical arcing.

Location: Typically used outdoors in rural or suburban areas where land is inexpensive and plentiful.

Pros: Lower initial capital cost; easier to expand or modify; components are easy to access for visual inspection.

Cons: Massive physical footprint; vulnerable to environmental factors (pollution, salt spray, lightning, and animals); requires frequent cleaning and maintenance.

4.1.2. Gas Insulated Substation (GIS)

A GIS uses Sulfur Hexafluoride (SF₆) gas as the insulating medium. SF₆ has a dielectric strength approximately 3 times higher than air, allowing the components to be placed much closer together.

Design: All active components (breakers, busbars, CTs/VTs) are encapsulated inside grounded metal tubes or "chambers" filled with pressurized SF₆ gas.

Location: Ideal for urban centers, offshore platforms, underground sites, or harsh industrial environments where space is at a premium.

Pros: Extremely compact (occupies only 10%–30% of the space of an AIS); nearly maintenance-free; immune to weather and pollution.

Cons: Significantly higher upfront cost (10%–40% more than AIS); SF₆ is a potent greenhouse gas requiring strict leak management; repairs are complex and require specialists.

4.2. Double Bus Double Breaker (DBDB)

The **Double Bus Double Breaker (DBDB)** arrangement is the ultimate redundancy configuration for high-voltage substations (typically 400kV and above). It ensures that no single point of failure—whether a bus fault or a breaker failure—will result in a total loss of power to a feeder.

1. The Architecture

In this setup, there are two independent main buses (Bus 1 and Bus 2). Every incoming or outgoing feeder is connected to both buses.

- Duplicate Hardware: Each circuit has two dedicated circuit breakers and four sets of isolators (disconnect switches).
- Parallel Paths: Under normal operating conditions, both breakers are closed. This means power flows through both paths simultaneously.

2. Operational Scenarios & Maintenance

The true value of DBDB is seen when things go wrong or require maintenance:

A. Breaker Maintenance

If Breaker A (connected to Bus 1) needs testing or repair:

1. The operator simply opens Breaker A.
2. The feeder remains energized through Breaker B (connected to Bus 2).
3. Result: Zero downtime for the load, and no complex "transfer" switching is required.

B. Bus Bar Fault

If a fault occurs on Bus 1 (e.g., an insulator flashover):

1. Bus 1 protection trips all breakers connected to it.
2. All circuits continue to be fed by Bus 2 through their second set of breakers.
3. Result: The substation remains fully operational despite a major bus failure.

C. Breaker Failure

If a circuit breaker fails to trip during a feeder fault:

1. The protection system will trip all other breakers on that specific bus to clear the fault.
2. However, the "healthy" bus continues to supply the feeders via their second breakers.

Result: Only the bus connected to the failed breaker is lost; the actual power flow to the feeders is maintained.

At starting of plant, the grid power or emergency diesel generator (EDG) runs the synchronous generator as a synchronous motor that starts the GT. When the achieved speed is at around 60% to 63% of nominal speed, the motor then starts to act as a generator. When the turbine speed reaches around 3%, firing begins in the combustion chamber. Generally, it takes around 18-20 minutes to start the gas turbine.

The gas turbine shaft connected to the generator is used to start generation. Bearing is used in gas turbine to support and allow smooth rotation between turbine shaft and the generator. This reduces friction and increases efficiency and stability.

While shutting down the gas turbine we cannot abruptly decrease the speed of the shaft, else temperature will increase due to friction resulting in expansion of the metallic body of the turbine blade. For this reason, turning gear is used to decrease speed gradually. Finally, a constant speed of 120rpm (~2% of rated speed) is maintained through turning gear which is called cranking speed or barring speed and excess heat is dissipated.

5.5. Heat Recovery Steam Generation (HRSG) System

The Condensate Extraction Pump (CEP) pumps condensate demi water from condenser to the economizer and deaerator towards the Boiler Feed Pump (BFP). BFP then feeds this condensate water to the Heat Recovery Steam Generation System (HRSG). There are two BFPs, one of which is on standby.

The flue gas generated from the GT is passed through a set of ducts which are carrying the demineralized water pumped by BFP. The BFP pumps water using 11 stages of pressure increasing mechanical structure to ensure approximately around 120 bar at HP container, 31 bar at IP container and 3 bar at LP container. High Pressure (HP) ducts are structurally carrying high temperature and high pressurized steam towards HP container. The next set of ducts are usually carrying less temperature and pressure steam towards Intermediate pressure (IP) container. The last set of ducts carrying lowest pressurized and temperature steam towards lower pressure (LP) container. Following Rankine cycle the steam generation is done. To ensure maximum work done super heating steam is generated.

At first, HP economizer increases the temperature of the demineralized water up to boiling point and sent to HP drum.

HP evaporator provides the latent heat to generate saturated steam.

HP super heater increases the temperature of the steam up to flue gas temperature. HP steam turns the HP turbine at around 120 bar and losses pressure that falls around 33 bar.

(Note: The 225MW plant requires HP pressure of 80bar.)

This steam and the excess HP steam is driven to cold reheat boiler.

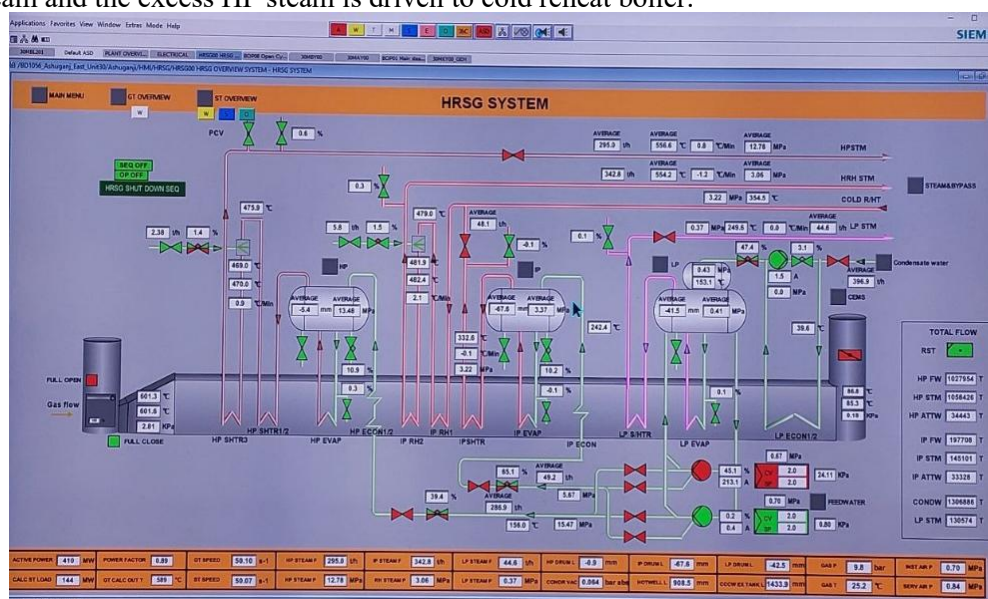


Fig. 3: HRSG System

IP economizer increases the temperature of the demineralized water up to boiling point and sent to IP drum.

IP evaporator provides the latent heat to generate steam.

IP super heater increases the temperature of the steam up to 333°C. IP superheated steam and cold reheated steam are then reheated using IP reheater 1 and IP reheater 2 to generate hot reheat (HRH) steam.

HRH steam then turns IP turbine. After losing temperature the steam and the excess HRH steam is driven towards to condenser.

Condensate water passes through the LP economizer and then deaerator removes entrapped gases, like oxygen and carbon dioxide and stored at LP drum.

LP evaporator generates steam which is superheated along with feedwater to generate LP steam.

LP steam drives the LP turbine. After losing the temperature LP steam and excess LP steam is driven towards condenser.

This steam and the excess HP steam is driven to cold reheat boiler.

The flue gas temperature falls at around 86°C and is released to atmosphere through chimney. Some remaining steam is released through vents to atmosphere.

5.6. Steam Turbine

Steam turbine consists of High Pressure (HP) Turbine, Intermediate pressure turbine and Low Pressure (LP) Turbine. This CCPP is designed for a single shaft generator. So, GT and ST are connected to a single generator on a single shaft. A Self Shifting Synchronization Clutch (SSSC) is used here to connect the ST with the generator.

Steam turbine can be started following any of the 3 methods depending of the temperature of the body and the shaft and how long the turbine was shut down.

- Hot start up (around 2 hours is required)
- Warm startup
- Cold startup

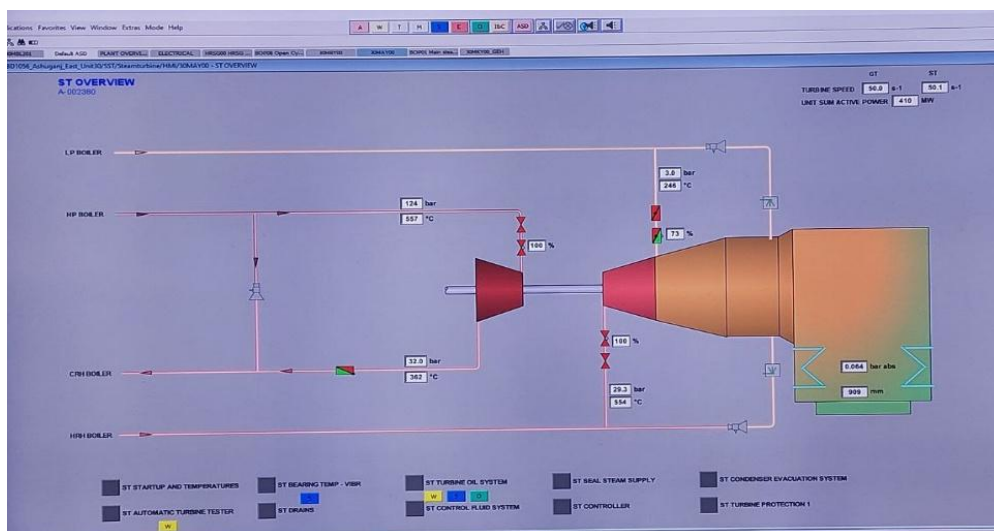


Fig. 4: Steam Turbine Overview

Different startup methods follow different startup characteristics curves.

Due to the maintenance work or some other reason, the power plant produces mechanical power by GT only. In that scenario, diverter damper is located between the GT and HRSG to evacuate the exhaust flue gas. The SSSC engages the ST or disengages when required.

5.7. Transformer

The Transformer placed in North CCPP to supply generated power to 400kV bus is an oil natural air forced (ONAF) transformer. The generator step-up transformer (GSUT) steps up the generated 22kV output voltage of generator to 400kV and connects the 400kV transmission line through gas insulated substation (GIS). There is another transformer that utilizes power to NORTH CCPP facility is called unit auxiliary transformer (UAT). The UAT is 22kV/6.6kV rated. The UAT ensures the 5% of generated power to deliver to run the facility itself. This value ensures the PGCB data of generated power of 393MW while the NORTH CCPP actually generating about 410MW. To emphasize the current condition the facility can generate even more power but that won't be necessary since the APSCL is contributing to grid on load sharing procedure. The grid auxiliary transformer (GAT) brings power from 230kV grid through GIS and supplies 6.6kV North CCPP facility to start up the plant from shutdown. Another transformer is supplying power to 225MW plant if the EAST plant requires start up.

Some transformer might represent 20 BAT 10 which means the bus auxiliary transformer is rated at 20kV and 10MVA. Some transformer characteristics shows 10 BBT 10 that also means it is rated at 10kV and 10MVA.

The GSUT of NORTH CCPP has 27 taps and 18 cooling fans. The UAT and the GSUT both has resistor temperature detector (RTD) to identify if any fire occurrences happens and the control unit can initiate firefighting actuators to prevent further damage. The UAT featured breather device which filters the cooling oil.

5.8. Generator

The generator mechanical input power is provided by GT and ST. The mechanical input power helps to produce electrical output power using the principle of electromagnetic induction. The generator shaft rotates a rotor with a magnetic field. The rotor magnetic field cuts across the stator windings, inducing an AC voltage in the armature coil. Generator produces electrical power at 22 kV. The power supplied to the power grid using a Generator Step Up Transformer (GSUT) to 400kV bus. A Unit Auxiliary Transformer (UAT) is used to step down the 22 kV to 6.6 kV to run the auxiliary equipment of the power plant.

Another stepdown transformer is used to step the voltage level down to 400V for other electrical equipment.

The east project also provides power to start 225 MW CCPP when required.

The power produced from Gas Turbine is primarily consumed by compressor.

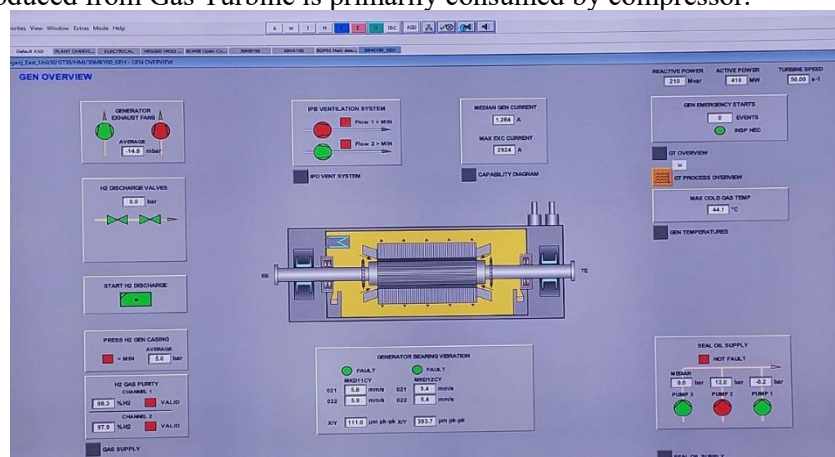


Fig. 5: Generator Overview

5.9. Condenser

A heat exchange device in a power plant that condenses the steam from the steam turbine back into water by cooling it with closed cycle cooling water. The treatment water is used here to produce steam. Conductance, pH level, temperature and dissolved mineral is maintained to produce the demineralized water. The close cycle cooling is being cooled down using open cycle cooling water. Here, River water is used as open cycle cooling water. The temperature difference of released open cycle cooling water and river is maintained below 5°C to reserve the bio-diversity. At the low-pressure terminal, we use a vacuum pump to create a negative pressure to prevent back flow of water in condenser.

6. Modes of Generation Control

6.1. Droop Characteristics

Load vs speed characteristics of a synchronous generator describes the generator's speed will decrease if load is added to the system but the speed will increase if load demand decreases. Droop control is essential when multiple generators run in parallel, allowing them to share the load and maintain a stable system frequency.

If the system is rated at 3000rpm which means the system can run at 3000rpm with full load condition. 5% droop characteristics suggests, the no load speed is at 105% level that of full load speed. System running at no load condition at 3150rpm, the speed would decrease down to 3000rpm rated speed if the load is added.

For 50Hz system, running with full load condition, would generate up to 52.5Hz system frequency if load is decreased.

6.2. FGMO Mode

The FGMO stands for frequency guided master oscillator that uses isochronous load sharing systems for parallel generators of GT and ST. In this design a master generator is defined that operates on isochronous mode and the other acts as slave generator which follows droops characteristics but their droop characteristics is guided by the master generator.

The master generator ensures the frequency of bus remains perfectly constant regardless of load changes while the slave generator stably share the load amongst themselves but the power output is constantly adjusted by a signal from the master to ensure the total generation matches the total load. Suppose G1 is master generator, G2 is slave generator. If the load increases the system frequency would start to drop. G1's controller sees this and immediately increases the fuel to G1 to bring the frequency back to optimum level. If load decreases, the G1 does the opposite.

To prevent G1 from hitting its maximum capacity while G2 would sit idle, the load sharing controller (LSC) is implemented.

LSC monitors the real power output of all generator and calculates the average load of slave and compares the master generator's load to a predefined set point. If, G1 is above its load set point, the LSC sends a signal to slaves telling them to raise their speed reference. This makes them appear faster to their governors and causing them to automatically take on more load. This is fully automated and continuous processing.

6.3. Load Hunting

Load hunting is an unstable condition, governor hunting, where a generator's engine speed and its output frequency and voltage oscillate continuously. Instead of maintaining a steady state, the engine rhythmically speeds up and slows down.

The engine sound repeatedly rises and falls and an observer can notice the sound goes high and low and is repeating periodically. The lights connected to generator pulse of flicker in brightness. The frequency and voltage also swing back and forth.

This occurs when engine is running and governor over compensates by injecting too much fuel causing speed level up a bit. The engine starts speeding up and governor detects higher speed than rated and reacts by cutting too much fuel. The engine speed falls below rated speed.

This cycle continues due to mechanical governor worn linkages or loose actuator lever. Incorrect transmitter value of proportion, integral and derivative (PID) controller that is tuned incorrectly. Fuel injection system fault can also cause such occurrence.

7. Power Plant Auxiliary System

Instrumentation and Control department balances the monitoring, operation and on field maintenance. Captive power system provides power to auxiliary machineries.

- ✓ Cooling System
- ✓ Valve System
- ✓ Blow Down System
- ✓ Fire Fighting System

7.1. Cooling System

7.1.1. Cooling System Using Hydrogen Gas

Hydrogen gas is used here to cool the rotor, stator and windings. Closed cycle cooling water is used here to cool down the hydrogen gas. Seal oil is used for partitioning the generator contained hydrogen gas from associated devices. If hydrogen still leaks out of generator chamber, an orange duct carries the necessary amount of hydrogen gas to inject using automated valve. This hydrogen is produced electrolysis of treatment water.

7.1.2. Cooling System Using Lubricating Oil

Transformer, bearing of turbine, generator, gear box, turning gear and other equipment are all being cooled down using lubricant. Lube oil pump is used to maintain the circulation of lubricant. Closed cycle cooling water is used to cool down the lubricant.

7.1.3. Cooling System Using Water

River water is demineralized to use in close cycle cooling system. Most of the devices of the system are being cooled using close cycle cooling water. River water is directly used as open cycle cooling water. We circulate this water in very few specific regions as it contains minerals and other heavy particles that might damage the devices in the long run.

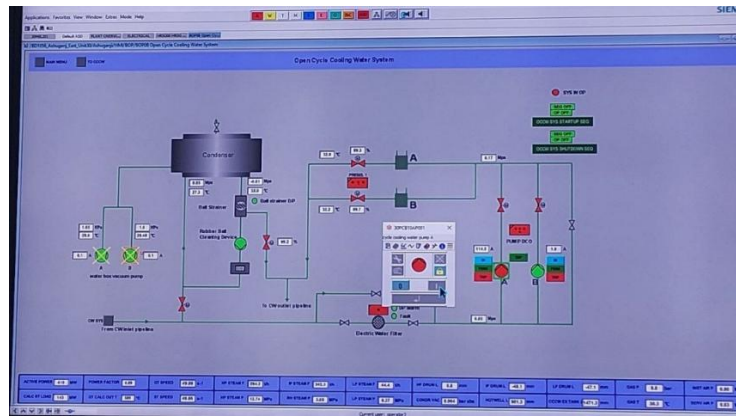


Fig. 7: Cooling System Overview

7.2. Valve System

Automated valves are used in many inlet and outlet points in almost every device of the system to regulate the circulation of fuel, air, water, lube oil and different gases as required. There are several types of valves such as:

7.2.1. Pneumatic Valve

To operate these types of valves instrumented air pressure is used. Instrumented air is a supply of compressed air that is treated to be clean, dry, and free of contaminants for use in industrial control systems and pneumatic instruments. Three air compressors are used to control the flow of instrumented air into the valves, two of which are on standby. Air pressure is maintained automatically by the system depending how much a valve should be opened. Generally, 5bar pressure is required to actuate pneumatic valve operation. Meanwhile the instrumented air pressure is up to 8bar.

7.2.2. Solenoid Valve

A solenoid valve is an electro-mechanically operated valve that controls the flow of liquids or gases by using an electric current to move a plunger with a magnetic coil. When the coil is energized, it creates a magnetic field that opens or closes the valve

7.2.3. Hydraulic Valve

A hydraulic valve properly directs the flow of a liquid medium, usually oil, through your hydraulic system. The direction of the oil flow is determined by the position of a spool. A hydraulic system can only function as per requirements by using valves.

7.3. Blowdown System

Despite being demineralized, condensate water might still contain some sludge and sedimented particles and minerals. Also, the chemicals in the condensate water might precipitate in the HP, IP and LP drums. This could potentially cause scaling, corrosion and foaming. Blowdown removes these contaminants and replaces them with treatment water. The precipitations are flushed down from the drums with a small amount of water. Cold water is added to it to cool it down. This mixture is then released to the river Meghna.

7.4. Fire-fighting System

A fire-fighting system is a safety device used in power plants and other industrial facilities to identify, manage, and put out fires. Fire detectors, alarms, sprinklers, hydrants, fire-water pumps, extinguishers, and specialized systems like CO₂ or foam suppression are typically included. In an emergency, the fire-water network is kept under pressure so that water can be supplied right away. In order to safeguard personnel, buildings, and equipment while stopping the fire from spreading, the system automatically sounds alarms and provides water or suppression agents when it detects a fire.

These auxiliary systems either run at 6.6kV or 400V. Gas Turbine Generator Starter Fuel Compressor runs at 6.6kV system. GTG SEE runs also at 6.6kV system. BFP, FF Water Pump, CEP Pump, Closed Cycle Water Pump, Natural Gas Booster Compressors runs at 6.6kV system.

Generally, 400V lines are providing power to the project facility of control, monitoring systems to execute operator controls.

8. Color Code

Let us know some ducts or channels passing through the facility and identify using color code.

The orange duct indicates it is carrying hydrogen gas to generator and thus the rotor and stator run in an air and moisture free environment to prevent rusting. It also helps cooling the chamber.

The red ducts are used for firefighting mechanism that carries water and it has actuators at different locations of a power plant.

The dark green ducts are used to carry demineralized (Demi water) water which is actually the condensate water or closed cycle water.

The light green ducts are carrying open cycle cooling water. And we know the OCW helps cooling of CCW and is being released to river within 5°C of temperature difference.

The yellow ducts are carrying lube oil to devices which is actually the cooling system.

The blue pipes are carrying instrument air to ensure valve operation.

9. A brief on how to start power generation after a scheduled shutdown

Some auxiliary devices are need to be operated all the time whether the plant is on shutdown mode or not. These auxiliary devices ensure instrumentation air supply, insulation to thermal abrasion, lubrications to machineries, prevention of rust etc. So, instrument air and valve operation air or gas pressure pumps, insulating and seal oil pumps, lube oil pumps are some of the auxiliary devices that are to be connected to power supply. In general, such jobs are ensured to be done by DC motors where power supply can be of energy storage of battery cells or rectified DC power from power grid.

If the power system was on a blackout, then, DC power storage is the only option to ensure such operational protocol.

The AC power from grid is connected to synchronous machine through Static Frequency Controller (SFC). SFC converts grid AC power to DC power and then generates variable AC power of variable frequency and voltage level.

Why variable frequency and voltage level?

This method ensures the synchronous machine is running at ~2Hz or ~120rpm using turning gear which is called barring speed or cranking speed so that the brushes of GT or generator don't get bruised if it starts running at rated speed. SFC ensures the frequency to be increasing gradually so that the

synchronous machine can be operated as a synchronous motor and gradually speed up the turbine speed to achieve rated speed.

If the system went on blackout, this scenario would be a lot worse because at that moment AC power supply from grid is barely producing enough power and also unstable to run such operation and requires highly skilled operational expertise. Some plants like 400MW North CCPP use Emergency Diesel Generator (EDG) to ensure the rotational speed of shaft.

SFC usually manages this procedure automatically and efficiently, leveraging contemporary technological advancements.

GT generates power from combustion of air and fuel or gas mixture. So, the compressor also turns alongside the GT and synchronous motor since they are all interconnected to a single shaft mechanism. The compressor brings compressed air to combustion chamber. The gas brought from Bakhraabad gas field channel, also gets purified and goes through Regulation and Metering System in gas processing facility of North CCPP project. Then Natural Gas Booster (NGB) supplies gas at optimal temperature and pressure. The electrical signal is sent to combustion chamber to start ignition. The number of burners starts increasing one by one. At this moment, the ignition can't produce much power to turn the GT. So, SFC is still needed to supply the power to shaft through motor that generates the torque. SFC gradually increases the shaft speed by supplying more power to motor which also shows increased frequency level. When the shaft reaches about 60% to 63% of the rated speed (around 1800rpm) the GT becomes self-sustaining. Generator Circuit Breaker breaks the power supply from grid and the synchronous machine is now operating as a generator since the GT already became self-sustained. But, the frequency and output voltage level of generator is still low. So, the governor mechanism ensures more gas flow to the combustion chamber which produces more pressure to GT blades. This pressure eventually supplies the extra torque requirements which ensures the GT acceleration. So, the GT speed starts increasing and at some point, the speed reaches the rated speed of 3000rpm.

So, the GT supplying the mechanical power to generate the electrical power at the generator output terminal. The exhaust is bypassed through the GT stack. At this point the efficiency of the power plant is about 30% to 35% or slightly higher.

But CCPP project is designed to be more efficient to regenerate power from the thermal energy. The diverter damper is then opened slightly so that the exhaust heat can't cause abrupt thermal shock to Heat Recovery Steam Generator (HRSG). The increasing amount of steam is passing through the HRSG system following the recommended standards of procedure. The boiler feed pump (BFP) is supplying power to HP, IP, and LP. At first HP superheating mechanism placed in HRSG stage cause superheating requires the maximum temperature and heat from exhaust flue gas for maximum pressure and temperature of HP steam. HP evaporator, HP economizer are placed successively. Then HRSG is designed to place IP reheater, IP superheater, IP evaporator, IP economizer, LP superheater, LP evaporator and LP economizer sequentially. The steam then helps spinning of HP turbine, IP turbine and ST turbine according to design. By this time the diverter damper is placed fully upwards and doesn't allow any exhaust to bypass to atmosphere but to only go through HRSG system in CCPP since the thermal shock prevention is already ensured.

Finally, the condenser condenses steam and sends it to condensate extraction pump CEP.

At the beginning of HRSG system, a vacuum pump is placed to help condense the steam so that the steam won't create back pressure.

The condensate water is subsequently treated and reintroduced by BFP into the HRSG system for steam generation, thereby sustaining the cycle. Some steam is always being released to atmosphere at temperature level of 86°C. But the facility always maintains the procedure of water treatment to add necessary amount of water.

10. Power Plant Protection

- ✓ Differential Protection
- ✓ Restricted Earth Fault Protection
- ✓ Directional Overcurrent Protection
- ✓ Directional Ground Fault Protection
- ✓ AC Re-closing Relay with SYN. Check
- ✓ Generator Auto-Synchronizing Relay
- ✓ Instantaneous Overcurrent Protection
- ✓ AC Time Overcurrent Protection
- ✓ AC Time Earth Fault Protection
- ✓ Compound Voltage Block Overcurrent Protection
- ✓ Breaker Failure Protection
- ✓ Reverse Phase Protection
- ✓ Transformer Thermal Protection
- ✓ Over-voltage Earth Fault Protection
- ✓ Over-excitation Protection
- ✓ Buchhaltz Protection
- ✓ Pressure Relief Protection
- ✓ Oil Level Protection
- ✓ Oil Temperature Protective Device
- ✓ Rotor Earth Fault Protection
- ✓ Stator Earth Fault Protection
- ✓ Out of Step Protection
- ✓ Reverse Power Protection
- ✓ Impedance Protection
- ✓ Frequency Protection
- ✓ Under-excitation Protection
- ✓ Under-voltage Protection
- ✓ Over-voltage Protection
- ✓ Inadvertent Energization Protection
- ✓ Field Breaker Protection
- ✓ Excitor Trouble Alarm
- ✓ AVR Failure Alarm
- ✓ ARC Protection

11. References

PGCB NOTE

https://docs.google.com/spreadsheets/d/1KnXIKOi0tJkr4fapaV-PKWQFA6XmX9ht/edit?usp=drive_link&oid=102169018295214007255&rtpof=true&sd=true

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